Public Wastewater Network Data in South Australia

What is openly available, whether it is usable, and what has to be requested

Optimal Sensor Placement Project · Data Availability Assessment · September 3, 2026

Scope. This report covers wastewater network data for the South Australian metropolitan utility network. It records what can be obtained publicly without credentials, assesses whether that data is good enough to build a network model from, and identifies what is missing and would have to be requested. Council-owned schemes outside the utility network are not catalogued here.

Every figure was verified by direct anonymous query and can be independently re-checked. No confidential or client-supplied material is reproduced. Section 6 names internal service paths and is marked for controlled circulation.

1 Purpose

Placing sensors on a wastewater network requires knowing the shape of the network: which pipes connect to which, which way water flows, how deep the pipes are, and where the access points are. Before any of that can be modelled, one has to establish what data exists and whether it is good enough to use.

A request for data access is far more likely to succeed when it names precisely what is wanted and can show that the freely available alternatives have already been exhausted. This report is intended to serve as that evidence.

2 Summary of findings

1. **The utility's own sewer network is partly public.** A council republishes the utility's sewer asset register for its own area: 1,002 gravity mains with invert levels on every record, and 360 maintenance holes as mapped points. This is the only public window onto metropolitan utility-owned sewer found (Section 3).
2. **Statewide gravity mains are openly downloadable.** A state service returns 201,713 gravity mains with levels, without credentials and without being clipped to a council boundary (Section 4). An earlier assessment recorded these layers as access-restricted; that was the map viewer's export control, not the query interface, and the finding is superseded.
3. **No public source anywhere records maintenance hole depth.** The utility's published manhole layer carries a usable surface level on only 10 of 360 records, and the statewide service has no manhole layer at all. Since maintenance holes are the set of candidate sensor locations, this is the most consequential gap in the public data.
4. **Depth can nonetheless be reconstructed, and the error is now measured.** Cover level minus invert level gives depth, and cover level can be interpolated from published contour lines to within about a quarter of a metre against chamber depths of roughly three metres (Section 5.1).
5. **No sensor readings for this network are public.** Instrument layers exist and are catalogued, but every one requires a login, and the publicly listed ones relate to the drinking water network rather than wastewater.
6. **One field convention must be verified before use, and is easy to get wrong.** The invert fields are labelled by flow rather than by geometry. Misreading them reverses the majority of pipes while leaving every record count unchanged (Section 3.2).

3 The utility network that is public

Publisher A local council, republishing the state water utility's sewer asset register for its own council area.

Address `https://services-ap1.arcgis.com/38lmAqtkEsaNGT6T/arcgis/rest/services`

Access Fully open. No credentials.

The wastewater-relevant layers, all under `SA_Water_Sewer_Network/FeatureServer` except where noted:

Content	Layer	Records
Gravity sewer mains	layer 6	1,002
Maintenance holes	layer 2	360
Property inspection points	layer 4	2,851
Sewer connections	layer 7	2,919
Inspection openings	layer 3	533
Rising mains (pressurised)	layer 8	7
All sewer point assets merged	`SA_Water_Sewer_Nodes/0`	3,761
Ground contours, 1 m steps	`Contours_1m/0`	6,944

The contour layer matters more than it looks. Depth of an access point is ground level minus the pipe invert level, so with contours supplying the first number and the pipe layer the second, depth can be worked out even though it is recorded nowhere in this data.

3.1 Attribute completeness, gravity mains

Measured across all 1,002 records. "Populated" counts values that are actually usable, so a field full of zero placeholders is reported as unusable rather than complete.

Field	Populated	Range observed	Assessment
`START_INVE`	100%	28.15 to 59.15 m	upstream invert, usable
`END_INVERT`	100%	26.96 to 58.33 m	downstream invert, usable
`GRADE`	100% (one zero)	0 to 9.61 %	usable
`NOMINALDIA`	100%	150 to 450 mm	usable
`FLOWDIRECT`	100%	coded 1 / 2 / 0	required, see below
`CONST_YEAR`	99.5%	1896 to 2017	age, usable for risk
`MATERIAL`	55%	mostly vitrified clay	456 records unknown
`ROUGHNESS`	0%	all zero	unusable, must be assumed

Invert level is the elevation of the inside bottom of the pipe, given as a height above a survey datum. **Complete invert levels are the significant result here**: they allow hydraulic gradient to be derived directly from the data rather than assumed.

Maintenance hole depth is effectively absent. Only 10 of 360 maintenance holes carry a usable surface level, and across the merged 3,761-record point layer only 35 do. Depth is therefore not directly available from this source.

3.2 How the invert fields are labelled

Querying for mains whose upstream invert exceeds their downstream invert (`START_INVE > END_INVERT`) returns 1,002 of 1,002, every record.

That result is a naming convention, not a measurement. These fields are labelled by *flow*, not by geometry: `START_INVE` is the upstream-of-flow invert whichever way the line happened to be digitised, and the flow-direction code says which geometric vertex that is. The inequality is true by definition and carries no information. It is **not** evidence that the network runs downhill, and **not** evidence that it cannot loop back on itself. Either property has to be demonstrated on an assembled graph instead.

This was established by measurement. At a chamber shared by two pipes, the invert the upstream pipe hands over must equal the invert the downstream pipe picks up, so the correct reading is the one that makes those agree. Three readings were tested against two independent publishers:

Interpretation	Utility network	Statewide layer
Inverts labelled by geometry	35.2% exact	32.6% exact
Direction inferred by comparing inverts	37.0% exact	36.4% exact
Inverts labelled by flow	**95.3%**	**85.0%**

The flow-labelled reading agrees to a mean of 4 mm; the alternatives disagree by 0.79 and 0.85 m. Reading it either other way silently reverses 583 of the 1,002 mains here and 1,867 of 3,336 on the statewide layer, which inverts every upstream and downstream conclusion while leaving record, node and edge counts unchanged. Under the correct reading, inconsistent chamber inverts fall from 462 nodes to 4.

What follows. The flow-direction field cannot be dismissed as a drafting artefact and skipped. It is the only thing that says which end of a pipe is which, and a dataset that lacks it, or carries it unpopulated, cannot be oriented from invert levels alone.

3.3 The dataset stops at the council boundary

It covers one council area, so any sewer continuing into a neighbouring council is cut where the boundary falls, mid-network. Serious analysis needs a whole drainage catchment, not an administrative area: where a dataset stops at that line, pipes that continue in reality appear to end, so a pipe that is really mid-network looks like the bottom of the system and the downstream consequences of a blockage there cannot be seen at all.

4 Statewide gravity mains

Address `https://agsplus.geodata.sa.gov.au/arcgis/rest/services/Infrastructure_Utilities/MapServer`

Access Fully open. No credentials. Pages normally on `resultOffset`.

This service resolves the boundary problem above: it is not clipped to an administrative area.

Layer	Content	Records
9	Waste Water Gravity Main	201,713
10	Waste Water Low Pressure	1,146
11	Waste Water Pumping	6,039
12	Waste Water Vacuum	491

Layer 9 carries invert levels at each end, nominal diameter, flow direction, material and status, with inverts populated on the large majority of records. It is the only source offering both catchment-complete extent and usable levels.

There is no maintenance hole layer on this service, nor on any other public service examined. Every wastewater layer is pipes. Since maintenance holes are the set of candidate sensor locations, a network taken from this source has to have its nodes derived by snapping pipe endpoints together, and carries no chamber depth at all.

5 What can be built from this

5.1 Depth can be reconstructed, with a measured error

Chamber depth is the attribute repeatedly identified above as missing, and it is required by any placement method that reasons about how far water backs up behind a blockage. Since depth is cover level minus invert level, and inverts are complete, the question is how accurately a cover level can be interpolated from published contour lines.

Table 1: Error in contour-interpolated cover level, against surveyed values.

Comparison	n	Mean abs	p90	RMSE
1 m contours vs the surveyed cover levels published on the utility manhole layer	10	0.23 m	0.41 m	0.29 m
5 m contours vs surveyed levels, external validation set	707	1.34 m	2.62 m	2.08 m
5 m contour subset vs the full 1 m layer, same terrain	1,009	1.85 m	4.54 m	2.59 m

The second and third rows measure the same quantity by different routes, namely what a coarser contour interval costs, and they agree at 1.34 and 1.85 m. That agreement is what makes the first row credible despite its small sample: with a 1 m interval the reconstruction lands within about a quarter of a metre, against chamber depths measured here at a median of 2.9 m.

The validation set. The middle row uses the one South Australian publisher found that records surveyed surface level, invert and depth in the same record. It is a council scheme rather than part of the utility network, and is cited here solely as the reference against which the reconstruction was measured, not as a usable data source for this work. Two screens were applied before measuring, both on independent grounds rather than on the error itself, which would be circular: records outside the contour layer's footprint were excluded as extrapolation, and records failing the publisher's own depth = surface − invert identity were excluded as corrupt. On the records that pass, that identity holds to **0.018 m** across 726 cases, which is independent evidence those values are sound.

The practical conclusion. Where a fine contour interval exists, depth need not be requested and can be reconstructed with a stated and modest error. Where it does not, the reconstruction degrades quickly: at a 5 m interval the error approaches half the quantity being estimated. Contour or terrain coverage, rather than the sewer data itself, is therefore the limiting factor over much of the state.

5.2 What still cannot be done

1. **Candidate sensor locations cannot be enumerated outside the one council area.** Maintenance holes are the only places a sensor can physically be installed, and no catchment-complete public source lists them.
2. **Roughness must be assumed, not read.** No public source carries usable roughness. Standard values by pipe material are the only option, and any result sensitive to it should be reported with that assumption stated.
3. **Nothing can be validated against events that actually happened.** No public source

records historical chokes or overflows, so any placement method can be argued for but not evidenced.

4. **Field labelling conventions must be verified rather than inferred**, as Section 3.2 shows. Misreading them is invisible to the usual sanity checks, because every record count stays identical.

On the utility's own published network, inverts are complete while maintenance hole depth and surface level are effectively absent. That pattern was checked against several other South Australian publishers and holds across all of them, so it is best read as a sector-wide characteristic of how wastewater asset records have historically been kept rather than as any one organisation's shortcoming.

6 What is missing, and what to request

Handling note. This section names internal service paths on utility-operated infrastructure. It is placed last, under its own heading, so it can be removed in a single cut if a version of this report is needed for wider circulation. Everything before this point is compiled from openly accessible sources and carries no such restriction.

No access was attempted beyond checking whether each address responds, and every one asked for a login. No credentials were used, offered, or sought. The addresses were obtained by reading publicly published interactive map definitions, which list the services they draw from. They were not obtained from any internal system.

6.1 Services that exist but require authentication

Service	Layers of interest	Why it matters
`sawgis.sawater.com.au /AquaMap/NetworkReticulation`	49 Maintenance Holes, 42 Gravity Main, 46 Connections, 52 Inspection Points	**Layer 49 is the key one.** It is the candidate sensor location set, and no public equivalent exists anywhere.
`sawgis.sawater.com.au /NetworkReticulation/Wastewater`	22 Gravity Mains, 35 Structures	Value has narrowed now that statewide mains are openly downloadable.
`maps.sawater.com.au /AssetWorksManagement/ServiceRequest`	3 Service Request (wastewater)	Geolocated customer-reported faults. The only route to validation data.

A public account on a commercial mapping platform (`https://services3.arcgis.com/ugMYGpcYkB7ZZz3l/arcgis/rest/services`) lists instrument layers whose names and descriptions are visible but whose data requires a login. This is direct evidence that sensor measurements are not publicly available. Those particular instruments are on the drinking water network rather than the wastewater network.

6.2 The request, in priority order

Ordered to lead with what is genuinely missing. Asking again for network geometry that can now be downloaded would weaken the rest of the request.

1. **Maintenance holes with levels** *(spatial data request, highest priority).* An extract of the wastewater maintenance hole layer for a defined study area, retaining surface level and depth attributes wherever recorded. No public source in South Australia publishes maintenance hole points with depths, so this cannot be substituted for. Service: `AquaMap/NetworkReticulation` layer 49.
2. **Incident history** *(operational data request, highest analytical value).* Historical service request, choke or overflow records for the same study area, with location and date. This is the only ask that would let a proposed method be tested against events that actually occurred. Service: `AssetWorksManagement/ServiceRequest` layer 3.
3. **Sensor data** *(clarification first, then request).* Confirmation of whether an instrument layer exists for the wastewater network, since the publicly catalogued ones relate to drinking water, and if so whether it holds sensor locations only or also time-series measurements. These are materially different datasets. Time series would additionally allow a detection threshold to be calibrated, which cannot be done from any public source.
4. **Network extract** *(lower priority than previously stated).* A gravity main extract covering a complete drainage catchment rather than a council boundary, retaining invert levels. Largely satisfied by the open statewide service in Section 4; worth requesting only for assurance of completeness and currency over a defined study area.

All figures were measured by direct query on the dates of the survey. Record counts and population rates on live services change as their publishers update them, and should be re-measured before being relied upon in a formal submission. Endpoint addresses and layer numbers are subject to change by their operators; layer descriptions are reproduced as labelled in the published map definitions and have not been independently verified, since the underlying services are not publicly readable.